

MAC Sub-Layer (2)

COMP90007 Internet Technologies

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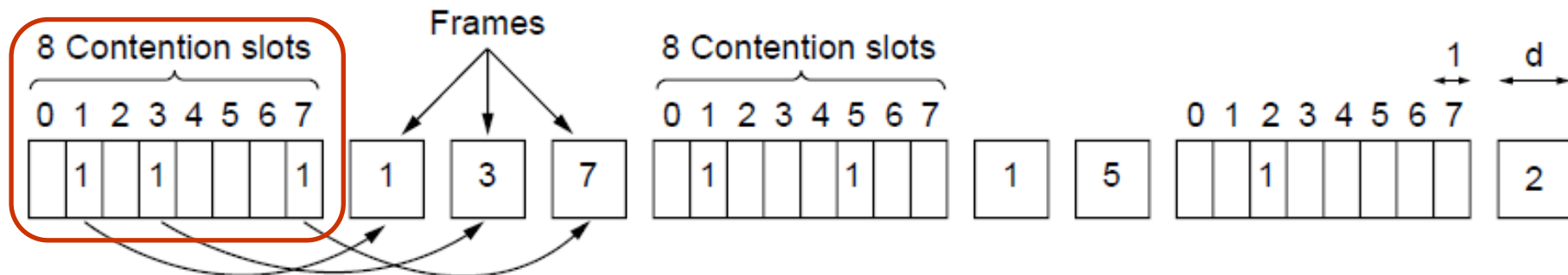
Multiple Access Protocols

- Contention
 - ALOHA, Slotted ALOHA
 - Carrier Sense Multiple Access
- Collision-Free
- Limited Contention
- MACA/MACAW (for Wireless LANs)

Collision-Free Protocols (1)

■ Bit Map Protocol

- ❑ Reservation-based protocol
- ❑ Separates the collection of transmission rights from the transmission events, no collisions
- ❑ Contention slots: 1 bit per station



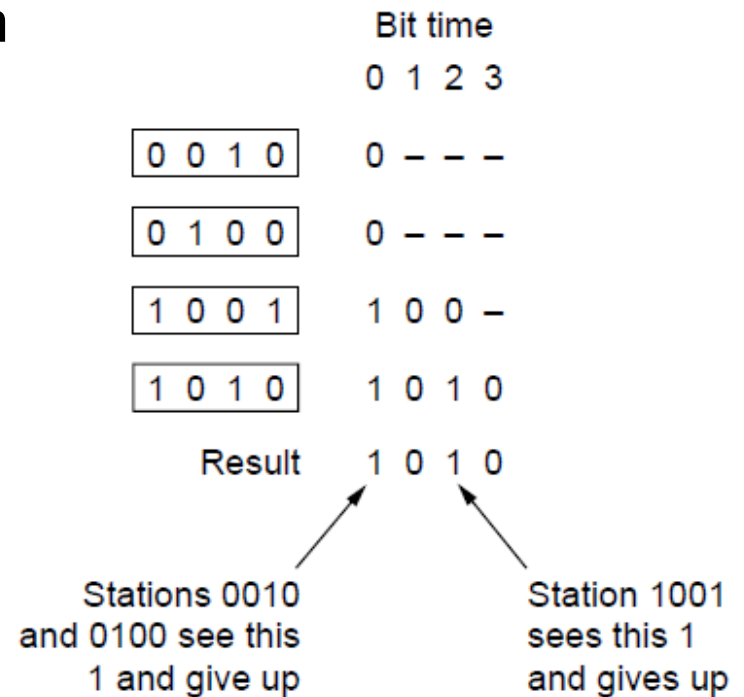
Collision-Free Protocols (2)

- Binary Countdown Protocol
 - Determines transmission order using binary station addresses
 - Stations with a higher binary address have a higher priority, no collisions

Collision-Free Protocols (3)

■ Binary Countdown Protocol

- ❑ **Stations** send their addresses from high-order bit during contention slots, $\log_2 N$ slots in total.
- ❑ **Channel** performs a bitwise OR operation; **stations** give up when they send a “0” but see a “1”
- ❑ After contention slots, the station that sees its full address is the next to send



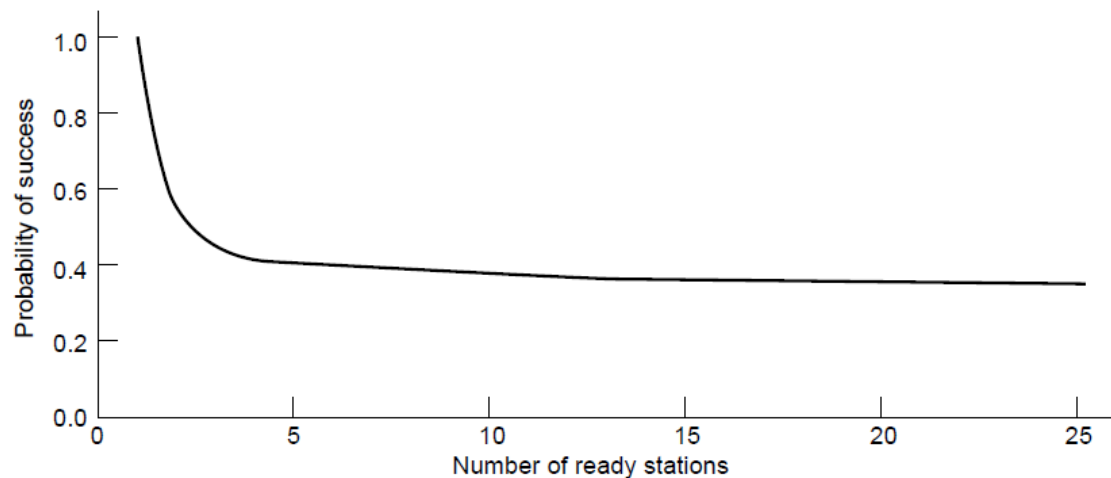
Contention vs. Collision-Free

■ Comparison

- Under **low loads** (collisions are rare), collision-free methods are less attractive due to the overhead.
- Under **higher loads**, contention methods are less attractive due to higher number of collisions.
- Both become inefficient at different points

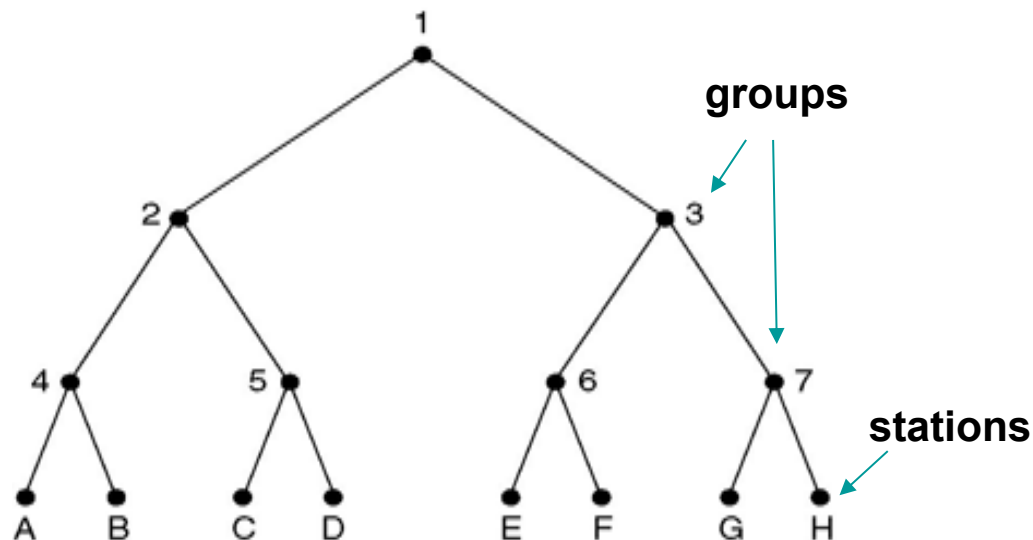
Limited Contention Protocols

- Contention model + collision-free model?
- Limited contention: divide stations into groups, within which only a very small number of stations are likely to transmit data
- Minimise the waste caused by idle or contention periods



Adaptive Tree Walk Protocol

- All stations compete for transmission, if a collision occurs, binary division is used to resolve contention
- Stations are divided into groups to poll
 - Perform **depth first search** over group nodes where collisions occur, i.e. when more than one station wants to transmit



Example 1: D G

Slot 1 → D, G – collision

Slot 2 → D

Slot 3 → G

Example 2: B D G

Slot 1 → B, D, G – collision

Slot 2 → B, D - collision

Slot 3 → B

Slot 4 → D

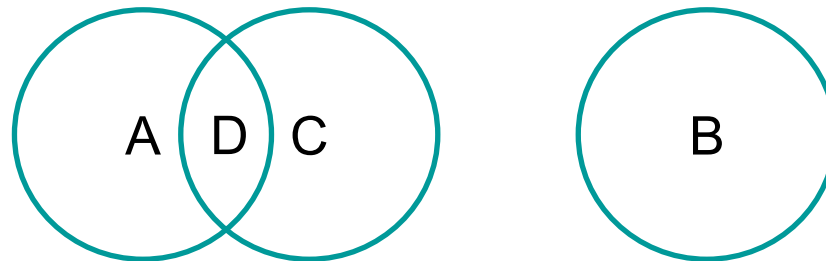
Slot 5 → G

Multiple Access Protocols

- Contention
 - ALOHA, Slotted ALOHA
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- **MACA/MACAW (for Wireless LANs)**

Wireless LAN Protocols

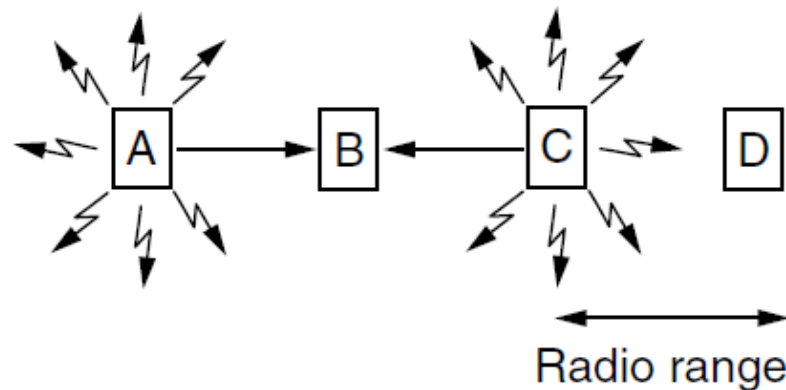
- Wireless complications: stations have limited coverage regions. When a station is within the range of two transmitters, interference can affect signal reception.



- **Hidden and exposed terminal problems**
 - Requires **detection** of transmissions **around the receiver**, not just carrier sensing
- **Protocols:**
 - Transmission Protocols for Wireless LANs (802.11)
 - Multiple Access with Collision Avoidance for Wireless (MACAW)

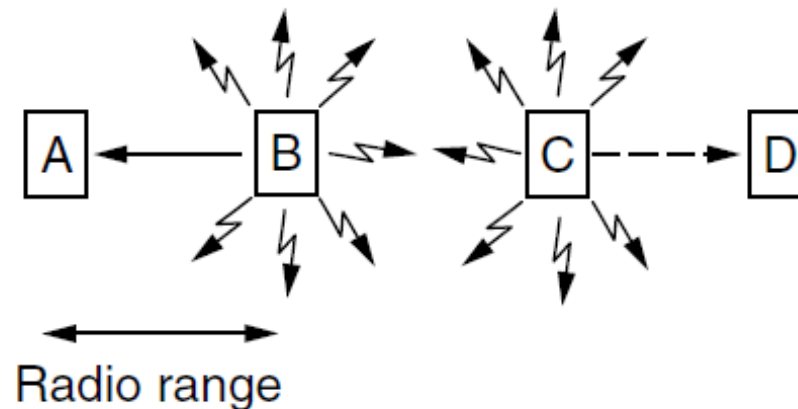
Hidden and Exposed Terminals (1)

- **Hidden terminals** are senders that cannot sense each other but still cause collisions at the intended receiver
 - A and C are hidden terminals when sending to B
 - Loss of efficiency; should be avoided



Hidden and Exposed Terminals (2)

- **Exposed terminals** are senders who can sense each other but still transmit safely to different receivers without causing interference
 - B and C are exposed terminals, when $B \rightarrow A$, $C \rightarrow D$
 - Desirable concurrency that improves performance

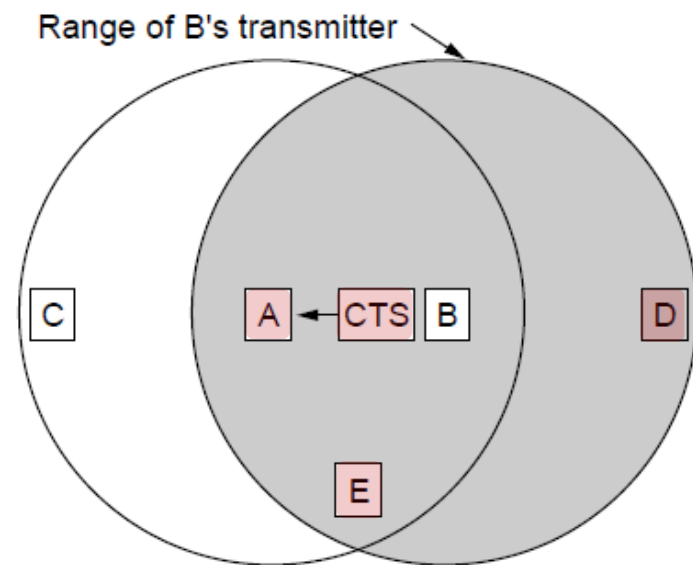
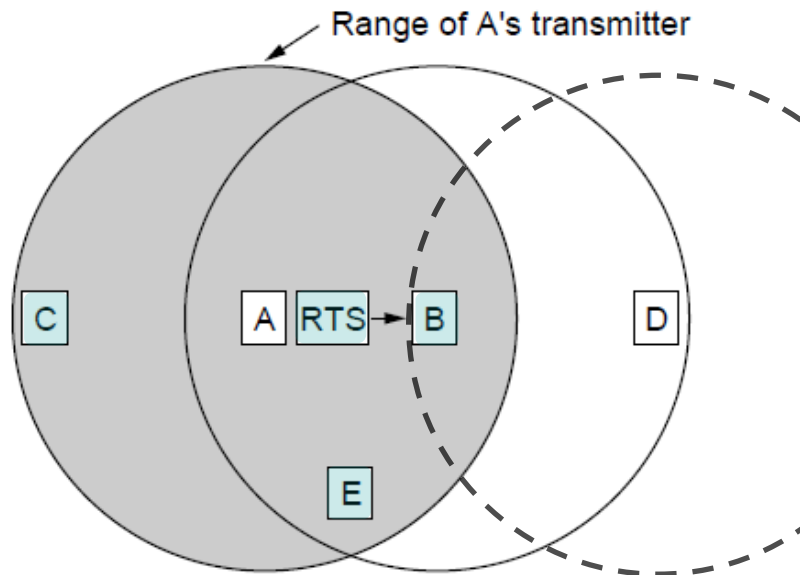


MACA

- MACA: Multiple Access with Collision Avoidance
 - The sender transmits an RTS, asking the receiver to respond with a short control frame CTS
 - All stations near the receiver hear CTS
 - The sender can then transmit data to the receiver
 - RTS = Request-to-Send; CTS = Clear-to-Send

MACA Example

- A wants to send data to B
 - **RTS** = Request-to-Send; **CTS** = Clear-to-Send
 - A can send with exposed but no hidden terminals



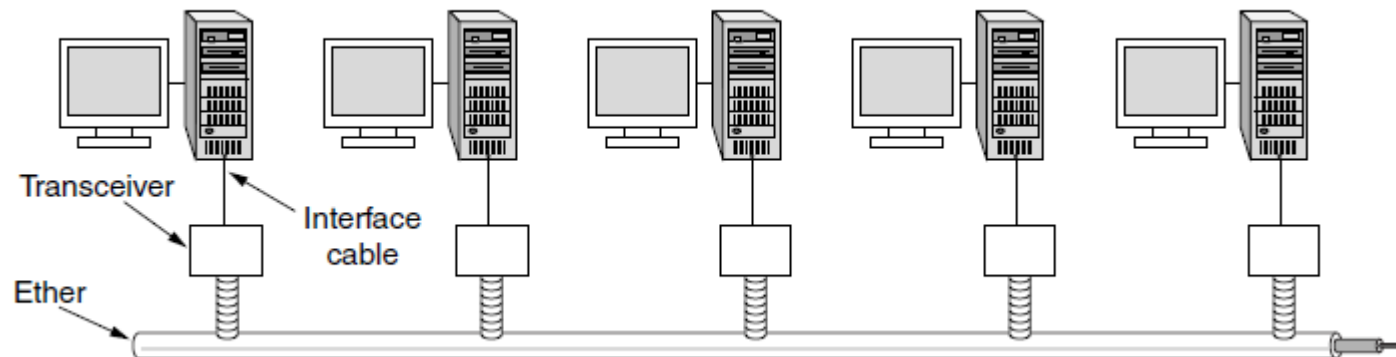
MAC Sub-Layer Case Study

■ Ethernet

- ❑ Classic Ethernet (3 ~ 10 Mbps)
- ❑ Switched Ethernet (100 Mbps, 1Gbps, 10 Gbps etc.)

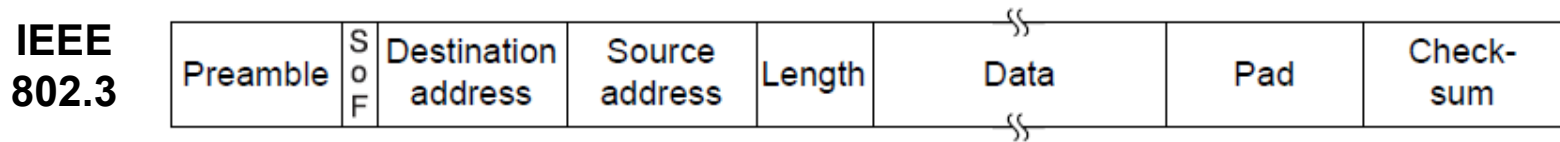
Classic Ethernet

- Each type of Ethernet has a maximum cable length per segment (e.g. 100m).
- Multiple segments can be connected by repeaters - a physical device which receives, amplifies and retransmits signals in both directions.



Ethernet Frame

- The frame format is still used in modern Ethernet



- **Preamble** (7B) – synchronisation between sender and receiver
 - **Start of Frame** (1B) – FLAG byte
 - **Dest. & Source addresses** (6B + 6B) – to identify sender and receiver
 - **Type or Length** (2B) – specifies which process to give the frame to
 - **Data** (0~1500B)
 - **Pad**(0~46B) – minimum Ethernet frame size 64 Bytes
 - **CRC** (4B) – 32 bits checksum
-
- MAC protocol: 1-persistent CSMA/CD
 - Random delay after collision is computed by BEB (Binary Exponential Backoff, with random number 0 to $2^i - 1$)

MAC Addressing

- The MAC Address provides a unique identifier for a physical network interface
- 48-bit number encoded in the frame, written in hexadecimal notation

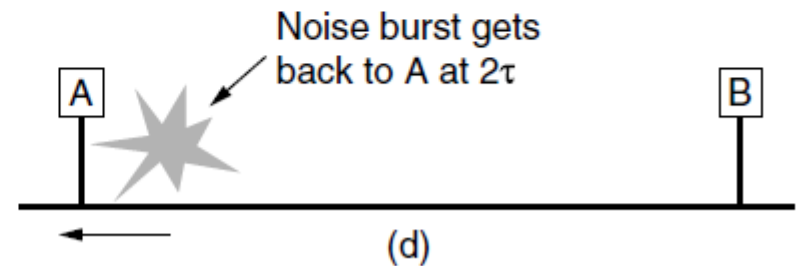
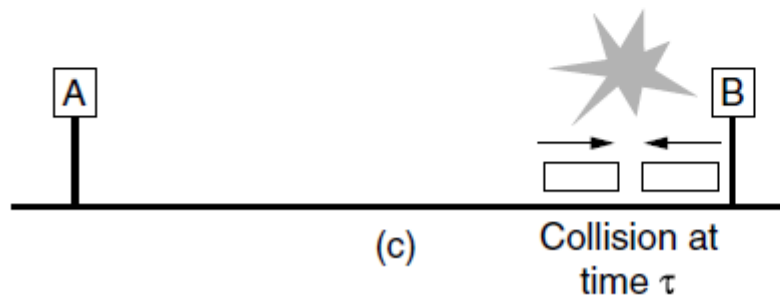
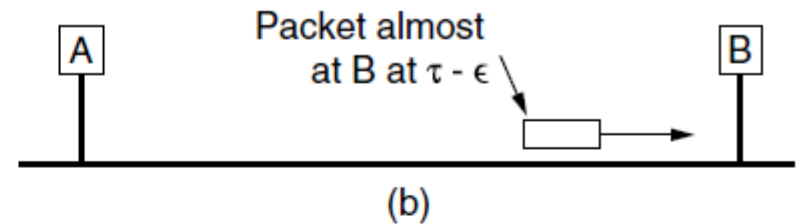
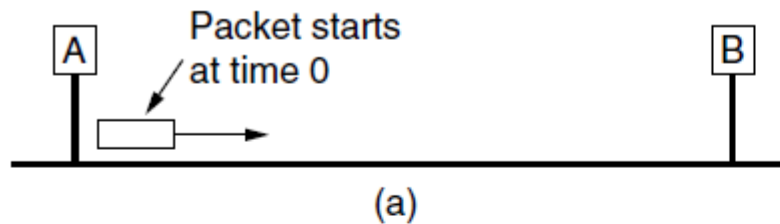
e.g. 00:02:2D:66:7C:2C

0	0000	4	0100	8	1000	C	1100
1	0001	5	0101	9	1001	D	1101
2	0010	6	0110	A	1010	E	1110
3	0011	7	0111	B	1011	F	1111

Hexadecimal to Binary Conversion Table

Classic Ethernet Minimum Frame Size

- Requires minimum frame size for reliable collision detection
 - Collisions can take as long as 2τ to detect
 - τ is the time taken to propagate over the Ethernet



Ethernet Performance

$$\text{Channel Efficiency} = \frac{1}{1 + (2BLE)/(cF)}$$

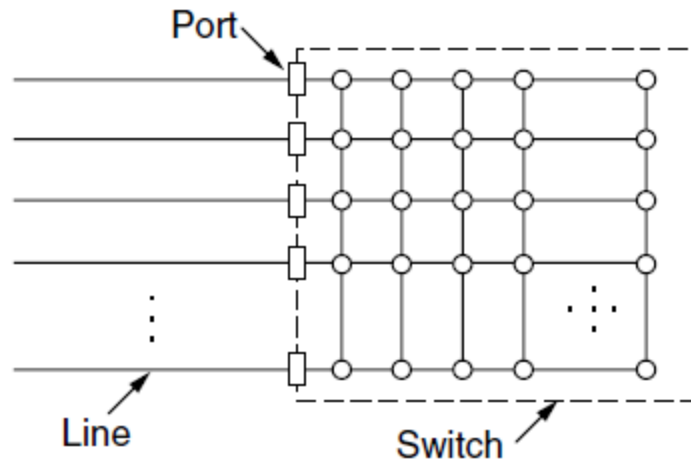
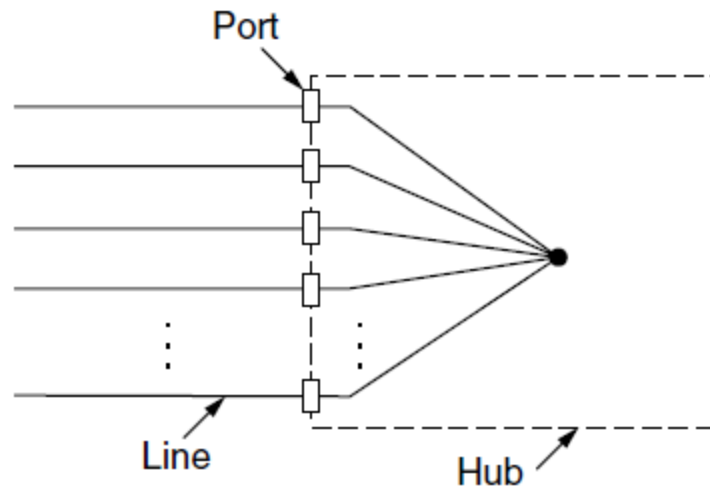
- ❑ F: frame length
 - ❑ B: bandwidth
 - ❑ L: cable length
 - ❑ c : speed of signal propagation; e : constant ≈ 2.71828
 - ❑ Optimal case: e contention slots per frame
-
- When cF is large, the channel efficiency is high.
 - Increasing network bandwidth (B) or distance (L) can reduce the efficiency for a given frame size.

Switched Ethernet



■ Hubs vs. Switches

- ❑ Hubs connect all devices into a single collision domain
- ❑ Switches isolate each port to a separate collision domain
 - Much greater throughput for multiple ports
 - No need for CSMA/CD with full-duplex lines



Summary

- Contention
 - ALOHA, Slotted ALOHA
 - Carrier Sense Multiple Access: 1-persistent, non-persistent, p-persistent
- Collision-Free: bit map, binary countdown
- Limited Contention: adaptive tree walk
- MACA/MACAW (for Wireless LANs)